

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

```
C:\Users\cdiks>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::7e77:2f70:f03e:47ea%14
    IPv4 Address. . . . . : 192.168.58.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.58.2
```

2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

```
C:\Users\cdiks>ping www.google.com

Pinging www.google.com [142.251.153.119] with 32 bytes of data:
Reply from 142.251.153.119: bytes=32 time=20ms TTL=128
Reply from 142.251.153.119: bytes=32 time=33ms TTL=128
Reply from 142.251.153.119: bytes=32 time=15ms TTL=128
Reply from 142.251.153.119: bytes=32 time=14ms TTL=128

Ping statistics for 142.251.153.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 33ms, Average = 20ms
```

3. Run the `tracert` (or `traceroute` on Linux/Mac) command to `www.flipkart.com` and list the number of hops it takes to reach the destination.

```
(kali@kali)-[~]
$ traceroute www.flipkart.com
traceroute to www.flipkart.com (103.243.32.90), 30 hops max, 60 byte packets
 1  192.168.58.2 (192.168.58.2)  1.084 ms  1.005 ms  0.934 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

4. Use the `nslookup` tool to find the IP address of `www.spotify.com` and explain in one line what this tool helps you verify.

```
C:\Users\cdiks>nslookup www.spotify.com
Server:      UnKnown
Address:     192.168.58.2

Name:       www.spotify.com.localdomain
Address:    10.0.0.123
```

5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

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    IPv4 Address. . . . . : 192.168.58.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.58.2

C:\Users\cdiks>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [23.55.244.185] with 32 bytes of data:
Reply from 23.55.244.185: bytes=32 time=14ms TTL=128
Reply from 23.55.244.185: bytes=32 time=15ms TTL=128
Reply from 23.55.244.185: bytes=32 time=14ms TTL=128
Reply from 23.55.244.185: bytes=32 time=17ms TTL=128

Ping statistics for 23.55.244.185:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 17ms, Average = 15ms

C:\Users\cdiks>tracert www.zomato.com

Tracing route to e70929.dsca.akamaiedge.net [23.55.244.185]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    192.168.58.2
  1  *         *         *         Request timed out.
  2  *         *         *         Request timed out.
  3  *         *         *         Request timed out.
  4  *         *         *         Request timed out.
  5  *         *         *         Request timed out.
  6  *         *         *         Request timed out.
  7  *         *         *         Request timed out.
  8  *         *         *         Request timed out.
  9  *         *         *         Request timed out.
 10 *         *         *         Request timed out.
 11 *         *         *         Request timed out.
 12 378 ms    244 ms    72 ms    a23-55-244-185.deploy.static.akamaitechnologies.com [23.55.244.185]

Trace complete.
```